Triple Integrals: Mass and Center of Mass of a Curved Solid

Stewart, Calculus: Early Transcendentals, Section 15.7, Exercise 40

Problem

Find the mass and the center of mass of the solid E with the given density function ρ. The solid E is bounded by the parabolic cylinder $z = 1 - y^2$ and the planes $x + z = 1$, $x = 0$, and $z = 0$. The density is $\rho(x, y, z) = 4$.

Symbols

- E is the solid, that is, the set of points (x, y, z) that satisfy the four defining inequalities.
- $z = 1 - y^2$ is a parabolic cylinder. Its surface does not contain the variable x. It is the same parabola in every plane $x = \text{const}$.
- $x + z = 1$, $x = 0$, and $z = 0$ are planes. Together with the cylinder they close the solid.
- $dV = dz\,dy\,dx$ is the volume element. The order of the factors shows the order of integration.
- $\rho(x, y, z)$ is the density, that is, mass per unit volume. Here $\rho = 4$ is constant.
- m is the total mass, $m = \iiint_E \rho\,dV$.
- $M_{yz} = \iiint_E x\rho\,dV$, $M_{xz} = \iiint_E y\rho\,dV$, and $M_{xy} = \iiint_E z\rho\,dV$ are the first moments of mass about the planes $x = 0$, $y = 0$, and $z = 0$.
- $\bar{x} = M_{yz}/m$, $\bar{y} = M_{xz}/m$, and $\bar{z} = M_{xy}/m$ are the coordinates of the center of mass.

The center of mass is the point where the solid balances. Each coordinate is a mass-weighted average of that coordinate over the solid.

Part (a): mass and center of mass

Step 1: describe the solid

Start with the cylinder bound $0 \le z \le 1 - y^2$. The inequality $1 - y^2 \ge 0$ gives $-1 \le y \le 1$. For each fixed pair (y, z), the plane $x + z = 1$ and the plane $x = 0$ give

$$0 \le x \le 1 - z. \tag{1}$$

Therefore

$$E = \{(x, y, z) : -1 \le y \le 1,\ 0 \le z \le 1 - y^2,\ 0 \le x \le 1 - z\}. \tag{2}$$

Two facts control all four integrals:

- The largest value of z is $1 - y^2$, not 1. At $y = 0$ the plane $x + z = 1$ and the cylinder meet at the apex $(0, 0, 1)$.
- For a fixed y, the section of E in the xz-plane is the trapezoid with vertices $(0, 0)$, $(1, 0)$, $(y^2,\ 1 - y^2)$, and $(0,\ 1 - y^2)$. Its top edge has length y^2.

Figure 1 shows the solid. Figure 2 shows the yz-section that gives the y-range and the xz-section that gives the x-limits.

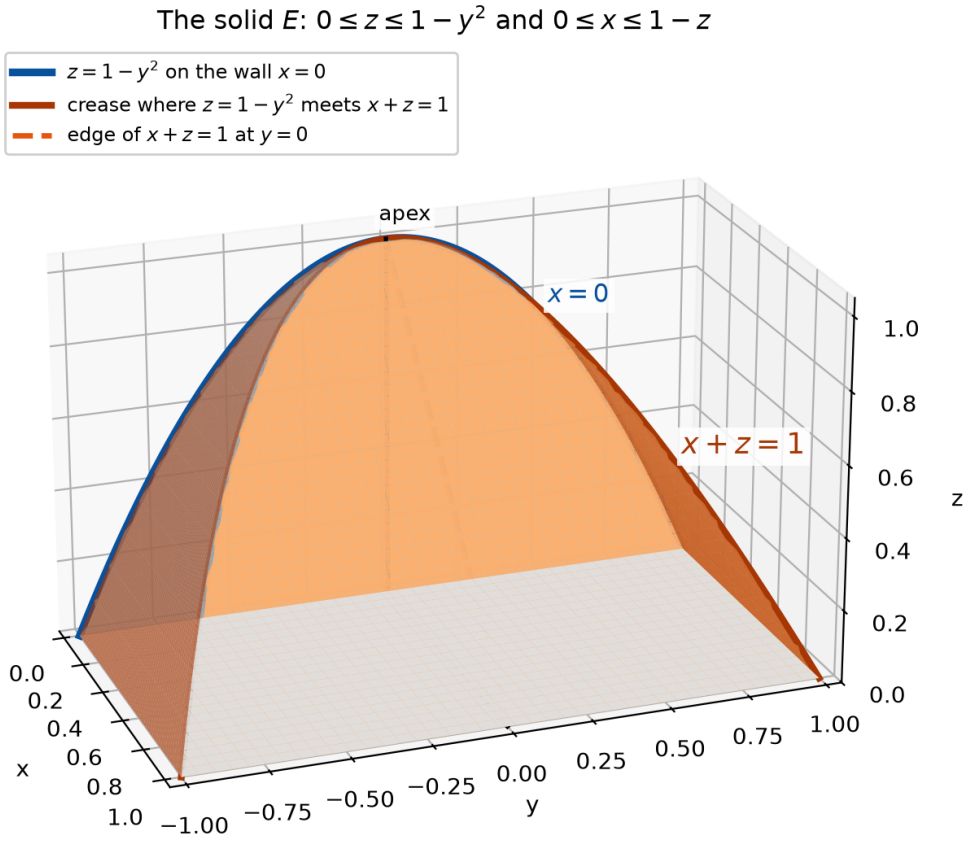


Figure 1: The solid E for Exercise 40. The orange face is the plane $x + z = 1$. The blue curve is the parabola $z = 1 - y^2$ on the wall $x = 0$. The dark curve is the crease where the cylinder meets the plane. The solid narrows to the apex $(0, 0, 1)$. The dark red surface is the part of the cylinder above the plane.

Step 2: choose the order of integration

Use the order x, then z, then y. The inner limits for x depend on z only. The middle limits for z depend on y only. The outer limits for y are constants. Each level uses only variables that are still free, so the order is legal.

The integrand in every case is a polynomial. No singularity lies in E.

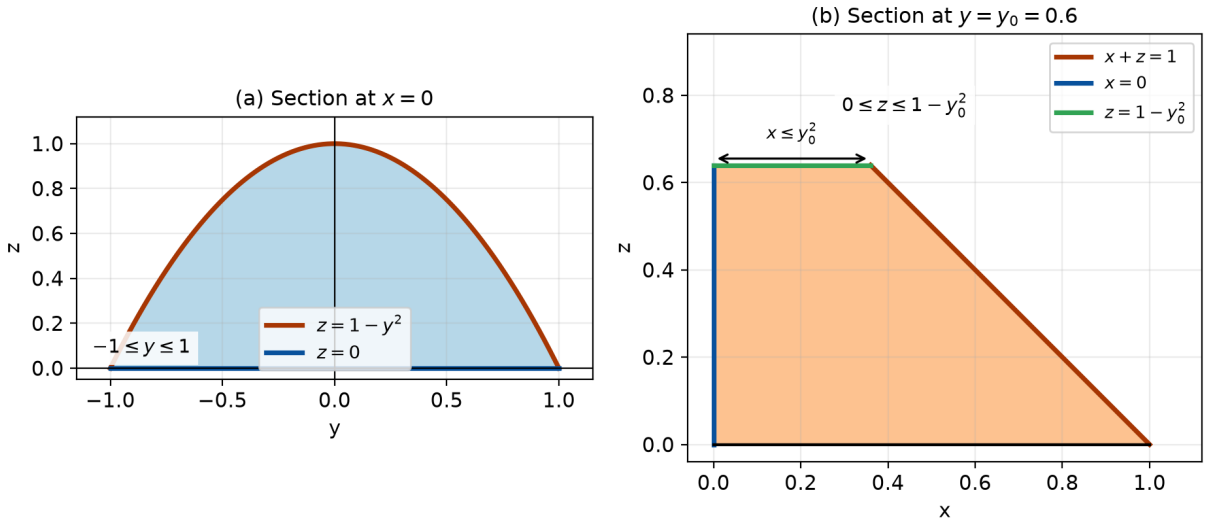


Figure 2: Left: the section at $x = 0$. The parabola and the line $z = 0$ give $-1 \le y \le 1$ and $0 \le z \le 1 - y^2$. Right: the section at $y = 0.6$. The section is a trapezoid with top edge from $x = 0$ to $x = y^2 = 0.36$ and slanted edge $x + z = 1$.

Step 3: compute the mass

$$m = \int_{-1}^{1} \int_{0}^{1-y^2} \int_{0}^{1-z} 4 \, dx \, dz \, dy. \tag{3}$$

The inner integral is

$$\int_{0}^{1-z} 4 \, dx = 4(1 - z). \tag{4}$$

The middle integral is

$$\begin{aligned} \int_{0}^{1-y^2} 4(1 - z) \, dz &= \Big[4z - 2z^2\Big]_{0}^{1-y^2} \\ &= 4(1 - y^2) - 2(1 - y^2)^2 = 2(1 - y^2)(1 + y^2) = 2(1 - y^4). \end{aligned} \tag{5}$$

The outer integral is

$$m = \int_{-1}^{1} 2(1 - y^4) \, dy = 2\left(2 - \frac{2}{5}\right) = \frac{16}{5}. \tag{6}$$

Thus $m = 16/5 = 3.2$. The volume is $V = m/\rho = 4/5 = 0.8$.

Step 4: compute the first moments

Moment about the plane $x = 0$.

$$M_{yz} = \int_{-1}^{1}\int_{0}^{1-y^2}\int_{0}^{1-z} 4x\,dx\,dz\,dy,$$

$$\int_{0}^{1-z} 4x\,dx = 2(1-z)^2, \tag{7}$$

$$\int_{0}^{1-y^2} 2(1-z)^2\,dz = \Big[-\frac{2}{3}(1-z)^3\Big]_{0}^{1-y^2} = \frac{2}{3}\left(1-y^6\right), \tag{8}$$

$$M_{yz} = \int_{-1}^{1}\frac{2}{3}(1-y^6)\,dy = \frac{2}{3}\left(2-\frac{2}{7}\right) = \frac{2}{3}\cdot\frac{12}{7} = \frac{8}{7}. \tag{9}$$

So $M_{yz} = 8/7 \approx 1.1429$.

Moment about the plane $y = 0$. The region E is symmetric about the plane $y = 0$: the change $y \to -y$ leaves every inequality unchanged. The integrand $4y$ is odd in y. Therefore

$$M_{xz} = \int_{-1}^{1}\int_{0}^{1-y^2}\int_{0}^{1-z} 4y\,dx\,dz\,dy = 0. \tag{10}$$

It follows that $\bar{y} = 0$. This is a symmetry argument, not an approximation.

Moment about the plane $z = 0$.

$$\int_{0}^{1-z} 4z\,dx = 4z(1-z), \tag{11}$$

$$\int_{0}^{1-y^2} 4z(1-z)\,dz = \Big[2z^2 - \frac{4}{3}z^3\Big]_{0}^{1-y^2} = 2\left(1-y^2\right)^2 - \frac{4}{3}\left(1-y^2\right)^3. \tag{12}$$

Write $A = 1 - y^2$. The last expression is $2A^2 - \frac{4}{3}A^3$. The two needed integrals are

$$\int_{-1}^{1} A^2\,dy = \int_{-1}^{1}(1-y^2)^2\,dy = \frac{16}{15}, \qquad \int_{-1}^{1} A^3\,dy = \int_{-1}^{1}(1-y^2)^3\,dy = \frac{32}{35}. \tag{13}$$

Hence

$$M_{xy} = 2\cdot\frac{16}{15} - \frac{4}{3}\cdot\frac{32}{35} = \frac{32}{15} - \frac{128}{105} = \frac{224-128}{105} = \frac{96}{105} = \frac{32}{35}. \tag{14}$$

So $M_{xy} = 32/35 \approx 0.9143$.

Step 5: divide each moment by the mass

$$\bar{x} = \frac{M_{yz}}{m} = \frac{8/7}{16/5} = \frac{8}{7}\cdot\frac{5}{16} = \frac{5}{14} \approx 0.3571, \tag{15}$$

$$\bar{y} = \frac{M_{xz}}{m} = 0, \tag{16}$$

$$\bar{z} = \frac{M_{xy}}{m} = \frac{32/35}{16/5} = \frac{32}{35}\cdot\frac{5}{16} = \frac{2}{7} \approx 0.2857. \tag{17}$$

Step 6: consistency checks

First, the point (5/14, 0, 2/7) lies in E. At the center, $1 - z = 5/7 = 0.714$, and $\bar{x} = 5/14 < 5/7$. Also $z = 2/7 < 1 - y^2 = 1$ at $y = 0$.

Second, the moments reproduce the mass times the coordinates:

$$\rho V \bar{x} = 4 \cdot \frac{4}{5} \cdot \frac{5}{14} = \frac{8}{7} = M_{yz}, \tag{18}$$

$$\rho V \bar{y} = 0 = M_{xz}, \tag{19}$$

$$\rho V \bar{z} = 4 \cdot \frac{4}{5} \cdot \frac{2}{7} = \frac{32}{35} = M_{xy}. \tag{20}$$

Both checks hold exactly.

Numerical confirmation

A Monte Carlo average with 4×10^6 points in the box $[0,1] \times [-1,1] \times [0,1]$ and a midpoint Riemann sum with $160^3 = 4.096 \times 10^6$ cells give the values in Table 1. The Monte Carlo relative error is at most 1.1×10^{-3} and the Riemann relative error is at most 2.8×10^{-3}.

Quantity	Exact value	Monte Carlo	Riemann sum
m	3.200000	3.200350	3.196961
M_{yz}	1.142857	1.142492	1.142569
M_{xz}	0.000000	−0.000646	0.000000
M_{xy}	0.914286	0.915112	0.911783

Table 1: Exact values and the two numerical estimates for Exercise 40.

The numerical moments give the centroid (0.35699, −0.00020, 0.28594), which agrees with (5/14, 0, 2/7).

An independent exact check by horizontal slices

A third exact route uses slices at constant z. The condition $z \le 1 - y^2$ is the same as $|y| \le \sqrt{1-z}$, so the slice of E at height z is the rectangle

$$0 \le x \le 1 - z, \qquad -\sqrt{1-z} \le y \le \sqrt{1-z}. \tag{21}$$

Its area is $A(z) = 2(1-z)^{3/2}$. Then

$$V = \int_0^1 2(1-z)^{3/2}\, dz = 2 \cdot \frac{2}{5} = \frac{4}{5}, \tag{22}$$

so $m = 4V = 16/5$. The same slices give

$$\bar{z} = \frac{1}{V} \int_0^1 z\, A(z)\, dz = \frac{5}{4} \cdot 2 \int_0^1 z(1-z)^{3/2}\, dz = \frac{5}{4} \cdot \frac{8}{35} = \frac{2}{7}. \tag{23}$$

This confirms m and $\bar{z}$ without any iterated integral.

Answer

$$\boxed{m = \tfrac{16}{5} = 3.2, \quad (\bar{x}, \bar{y}, \bar{z}) = \left(\tfrac{5}{14},\, 0,\, \tfrac{2}{7}\right) \approx (0.3571,\, 0,\, 0.2857)} \tag{24}$$

How to solve problems of this kind

1. Find the full range of each variable before you integrate. A cylinder bound such as $z \le 1 - y^2$ also bounds y, because a negative height is impossible.
2. Draw one section for each plane. The yz-section at $x = 0$ shows the y-range. The xz-section at fixed y shows the x- and z-limits.
3. Choose an order that keeps every limit simple. Here x first, then z, then y works, because the plane gives $x \le 1 - z$.
4. Use symmetry before you integrate. If the solid and the density do not change when $y \to -y$, then $M_{xz} = 0$ without any work.
5. Keep the density outside every inner integral when it is constant. This shortens the arithmetic.
6. Test the center of mass. It must lie inside the solid, and every moment must equal ρV times the matching coordinate.
7. Confirm the answer with a numerical method. A Riemann sum or a Monte Carlo average finds wrong limits quickly.

Summary of results

Quantity	Symbol	Value
density	ρ	4
volume	V	$4/5 = 0.8$
mass	m	$16/5 = 3.2$
moment about $x = 0$	M_{yz}	$8/7 \approx 1.1429$
moment about $y = 0$	M_{xz}	0
moment about $z = 0$	M_{xy}	$32/35 \approx 0.9143$
centroid x	$\bar{x}$	$5/14 \approx 0.3571$
centroid y	$\bar{y}$	0
centroid z	$\bar{z}$	$2/7 \approx 0.2857$

Table 2: All exact values for Exercise 40.